

Eric (Chen-Chien) Chang

BACKEND / PRODUCT ENGINEER

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PROFILE

Backend engineer who turns raw data and slow workflows into products people can use. Built systems that structure regulations, cut semiconductor simulation feedback from hours to minutes, and connect AI assistants to live real-world data.

EXPERIENCE

Lawsnote

Jan 2024 – Dec 2024

Taipei, Taiwan

Backend Software Engineer

- Streamed CPU-intensive analysis results, letting users see output in **5s instead of waiting 100s**.
- Added monitoring that caught missing source documents before they reached regulatory workflows.
- Removed a cross-language bottleneck in a Node.js native module through profiling.
- Converted ambiguous regulatory text into structured data and automated government-act refreshes.

MediaTek (Contract)

Feb 2022 – Mar 2023

Hsinchu, Taiwan

Backend Software Engineer

- Cut engineers' data-warehouse UI wait from **60s to 1s** by redesigning its PostgreSQL access path (case study).
- Cut simulation-data processing from **2hr to 1min**, shortening engineering feedback cycles.
- Kept long-running simulations reliable with a queue-backed service across a **5-node** cluster.
- Cut Go service memory use by **50%** and added telemetry to diagnose failures and resource pressure.

National Cheng Kung University

Sep 2017 – Jul 2019

Tainan, Taiwan

Research Assistant

- Conducted FDM 3D-printing research and built experimental tooling with Python, C, MicroPython, and Arduino.

SELECTED PRODUCT

Locus — Real-World Context Federation for AI Agents

2026 – Present

Cloudflare Workers / R2 / MCP

Independent Open-Source Product

- Built and launched a federation layer that gives AI clients one MCP connection to trustworthy live context across providers.
- Designed a provider-neutral contract for identity, location, freshness, expiry, provenance, confidence, coverage gaps, partial failures, and conflicting observations; backed it with scheduled TDX ingestion and R2 projections.
- Shipped source-backed decisions that combine available parking with compatible EV chargers across ChatGPT and text-only clients, explaining conflicts and requiring confirmation before relaxing live-availability constraints.

OPEN SOURCE LEADERSHIP & CONTRIBUTIONS

- Apache Kafka Contributor** — fixed offset commits during partition switching and JVM container resource detection (PR #21584); now contributing bounded buffering to Kafka Streams (PR #22458).
- Apache Gravitino Committer** — made PostgreSQL/MySQL integration tests reproducible, cut the Hive test image by **420MB**, and completed Python partition APIs.
- Apache DataFusion / Comet Contributor** — added Spark-compatible SQL behavior in Rust and connected it to Comet's native execution with end-to-end coverage.
- redis/rueidis Contributor** — built a compatibility adapter that lets teams migrate from go-redis with fewer application changes (PR #395).

TECHNICAL SKILLS

Languages: Go, Java, Rust, JavaScript / TypeScript, Python, C

Data & processing: PostgreSQL, ClickHouse, Hive, Kafka, Kafka Streams, Flink, Spark, SQL

Infrastructure: Airflow, Docker, Kubernetes, Linux, GitHub Actions

Observability: Prometheus, Grafana, OpenTelemetry, Pyroscope, Great Expectations

EDUCATION

National Taiwan Ocean University — Electrical Engineering coursework (Drop Out)

2013 – 2017